Starship V3 Reaches Space on First Flight, Despite Booster Crash and Engine Losses

From: Space.com Mined: 2026-05-23 5 min read

SpaceX's Starship program took a dramatic leap forward on Friday with the inaugural flight of Version 3 the most powerful iteration yet of the megarocket designed for Moon and Mars missions. The 408-foot-tall (124-meter) vehicle lifted off from a newly completed pad at Starbase, Texas, at 6:30 p.m. EDT, marking the 12th suborbital test flight overall and the first since October 2025.

The launch was not flawless. One of Super Heavy's 33 **Raptor engines** failed at liftoff, and the booster later missed a critical 'boostback' burn intended to guide its return. Instead of a controlled soft splashdown in the Gulf of Mexico, the massive first stage plummeted and crashed though within a pre-cleared safety zone. 'The booster didn't complete its full boost back,' said SpaceX spokesperson Dan Huot. 'Its mission ended a little bit early.'

Meanwhile, the upper stage Ship 39 lost one of its six engines during ascent but still reached space on the remaining five. 'I wouldn't call it **nominal orbital insertion**, but we're on a trajectory that we had analyzed, and it's within bounds,' Huot noted. The ship went on to perform a key objective: deploying 22 payloads via a '**PEZ dispenser**'-like door. Twenty were dummy Starlink satellites; two were actual Starlink spacecraft equipped with imaging sensors to inspect Starship's **heat shield tiles**.

Those cameras delivered a stunning view. As the final two simulators drifted away, SpaceX broadcast live video of Starship silhouetted against the blackness of space. 'That is a Starship in space,' Huot said, capturing the moment's significance.

Starship V3 represents a major design overhaul. Unlike its V2 predecessor, which used a disposable interstage ring, V3 integrates similar hardware directly onto the booster like a fence around the fuel tank dome to give the upper stage's engines room to ignite during '**hot staging**.' This mission also debuted a second launch pad, reducing risk to existing infrastructure.

Despite the booster loss, CEO Elon Musk celebrated on X: 'Congratulations SpaceX team on an epic first Starship V3 launch & landing! You scored a goal for humanity.' The test gathered crucial data on the new design's performance under stress, inching closer to operational missions that could one day carry astronauts to deep space.

Word Treasure

Raptor engines -- The powerful methane-fueled engines built by SpaceX to power the Starship and Super Heavy rocket.

PEZ dispenser -- A mechanism that ejects payloads one by one through a slot, similar to a candy dispenser.

nominal orbital insertion -- A perfectly on-target placement of a spacecraft into its intended orbit, without any deviations.

hot staging -- A risky technique where the upper stage ignites its engines while still attached to the booster, using special hardware to protect both stages.

heat shield tiles -- Protective ceramic plates that cover a spacecraft to prevent it from burning up during reentry into the atmosphere.

Background you need

SpaceX's Starship is the most powerful rocket ever built, intended to carry astronauts to the Moon and eventually to Mars. The company, founded by Elon Musk, uses a rapid trial-and-error philosophy, where each test flight -- even a failed one -- provides valuable data.

The V3 test was the 12th overall suborbital flight for the Starship program and the first major mission since October 2025. Earlier versions had mixed results, with some prototypes exploding during landing attempts before successful touchdowns were achieved. The 'hot staging' method debuted in a previous version and has now been made a permanent part of the design.

NASA's Artemis program plans to use a version of Starship as a lunar lander, so every test brings the U.S. closer to returning astronauts to the Moon.

Break it down

LEAD: Starship V3's first flight marks a dramatic leap forward, though not without serious hitches.

BOOSTER FAILURE:

+- 1 of 33 Raptor engines failed at liftoff.

+- Booster missed its boostback burn and crashed instead of a soft splashdown.

UPPER STAGE PERFORMANCE:

+- Ship 39 lost 1 of 6 engines but still reached space.

+- It deployed 22 payloads using a PEZ-dispenser-like door.

MILESTONE MOMENT: Live video showed Starship silhouetted against space -- a first for this configuration.

DESIGN EVOLUTION:

+- V3 integrates hot-staging hardware on the booster, eliminating the disposable interstage ring, and used a new pad to protect infrastructure.

CEO REACTION: Elon Musk celebrated the 'epic' launch, calling it a goal for humanity.

OUTLOOK: The test gathered crucial data, moving the program closer to operational deep-space missions.

Why it matters

Starship's success or failure directly affects the timeline for humanity's return to the Moon and eventual journeys to Mars. For teens today, these tests are the foundation of a future they may one day participate in -- as engineers, scientists, or even astronauts.

Test what you remember

1. **What happened to the Super Heavy booster during the V3 test flight?**
   - Ⓐ It landed safely in the Gulf of Mexico
   - Ⓑ It missed its boostback burn and crashed
   - Ⓒ It separated too early and crashed into the launch pad
   - Ⓓ It successfully completed a soft splashdown
2. **How many payloads did Ship 39 deploy?**
   - Ⓐ 6
   - Ⓑ 20
   - Ⓒ 22
   - Ⓓ 33
3. **What was special about the two real Starlink satellites onboard?**
   - Ⓐ They were the first Starlink satellites to orbit the Moon
   - Ⓑ They carried imaging sensors to inspect the heat shield
   - Ⓒ They tested a new laser communication system
   - Ⓓ They were designed to be retrieved mid-flight
4. **What design change does Starship V3 introduce compared to V2?**
   - Ⓐ A disposable interstage ring
   - Ⓑ Integrated hot-staging hardware directly on the booster
   - Ⓒ A larger payload bay with side doors
   - Ⓓ An escape tower for crew missions
5. **What phrase did SpaceX spokesperson Dan Huot repeatedly use during the launch?**
   - Ⓐ 'Mission accomplished'
   - Ⓑ 'We've had an anomaly'
   - Ⓒ 'I wouldn't call it nominal orbital insertion'
   - Ⓓ 'That is a Starship in space'
6. **Why was a second launch pad used for this mission?**
   - Ⓐ The first pad was damaged during a storm
   - Ⓑ To reduce risk to existing infrastructure
   - Ⓒ To fulfill a NASA requirement
   - Ⓓ To accommodate the taller V3 vehicle

Answer key (try the quiz first!): 1: B 2: C 3: B 4: B 5: D 6: B

Questions to think about

1. Positive: The flight successfully reached space, deployed cargo, and captured stunning video -- proving the new V3 design works. Every test inch the program closer to regular Moon and Mars missions, and even the failures teach important lessons.

2. Negative / critical: Losing multiple engines and crashing the booster raise concerns about whether the vehicle is reliable enough for crewed flight. Critics also point to the environmental impact of repeated launches and the risk that public excitement might mask real engineering problems.

3. Neutral / analytical: SpaceX operates on a 'fail fast, learn faster' model. The booster loss and engine issues are not disasters but expected steps in an iterative development process. This test mirrors earlier flights where final success came only after multiple partial failures.

4. Forward-looking: The next major hurdle is mastering booster recovery, a key to making Starship fully reusable. Future tests may attempt a tower catch like the Falcon 9 system. If successful, Starship could dramatically lower the cost of space access and reshape the entire space industry.

MY THOUGHTS (at least 20 words):